

SR AI ML ENGINEER

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Professional Summary

Senior AI Engineer with 5+ years of extensive experience productionizing enterprise Generative AI, autonomous multi-agent orchestration, low-latency microservices, and distributed ML pipelines. Expert in resolving system bottlenecks including context overflow, state drift, p99 retrieval latency spikes, and hallucinations across healthcare and banking environments. Specialist in zero-trust AI, hybrid RAG, fine-tuned LLMs, and MLOps infrastructure supporting high-concurrency workloads with 99.9% uptime SLAs.

Education

Master of Science in Computer and Information Science
The University of Kansas, USA

Aug 2023 – May 2025

Technical Skills

AI Architecture & Agents: Multi-Agent Orchestration, LangChain, LangGraph, Advanced RAG, Fine-Tuning (LoRA, QLoRA, PEFT), Semantic Caching, RAGAS, BERTScore

Backend & APIs: Python, FastAPI, Asynchronous Streaming, REST APIs, Microservices, OAuth 2.0, JWT, RBAC, Real-Time PII Masking

Databases & Search: PostgreSQL, Redis, FAISS, Pinecone, Weaviate, Apache Kafka, Hybrid Search, Cross-Encoder Reranking

ML & Frameworks: PyTorch, TensorFlow, Transformers, BERT, RoBERTa, Sentence-Transformers, NER, Text Classification, Scikit-Learn

Cloud & MLOps: AWS (SageMaker, Bedrock, Lambda, S3), Azure OpenAI, GCP Vertex AI, Docker, Kubernetes, Terraform, MLflow, Jenkins CI/CD, Databricks, Spark

Professional Experience

Cigna Health — Senior AI / ML Engineer (August 2025 – Present)

- Architected an autonomous multi-agent orchestration framework using Python, LangGraph, and Claude 3.5 Sonnet to parse 15+ complex medical benefit structures, eliminating structural hallucinations (dropping error rates from 18% to 7%) while boosting overall parsing accuracy to 94%.
- Built asynchronous FastAPI microservices backed by PostgreSQL read-replicas and Redis semantic caching to prevent connection pool exhaustion under heavy traffic, sustaining high-concurrency throughput up to 800 req/min peak parallel load.
- Engineered a dual-stage hybrid retrieval topology combining sparse keyword search, dense vector retrieval (Pinecone, FAISS), and Cross-Encoder reranking over 25TB+ of healthcare records—slashing p99 semantic search latency by 60% (from 3.2s down to 1.3s) at 92% retrieval precision.
- Implemented an active learning feedback loop using Azure Functions and MLflow to continuously capture edge-case parsing deviations during expert review, reducing downstream extraction failure rates by 23%.
- Designed a zero-trust streaming ingestion pipeline on Azure Blob Storage and Docker featuring real-time PII masking and RBAC controls, cutting false-positive security threat alerts by 40%.
- Developed high-throughput, asynchronous Java Spring Boot microservices integrated with Spring Cloud, PostgreSQL read-replicas, and Redis semantic caching, preventing connection pool exhaustion and sustaining parallel load up to 800 req/min.
- Built automated model evaluation gates in Jenkins CI/CD integrating RAGAS and BERTScore metrics to catch logical regression prior to production releases, yielding a 35% gain in response consistency.
- Implemented dynamic token reduction heuristics and intelligent model routing algorithms across high-volume inference calls, driving a 35% net monthly cost reduction in commercial LLM usage.
- Deployed resilient, containerized ML pipelines on Kubernetes using Terraform IaC, enforcing 99.9% production uptime SLAs while maintaining 85%+ Pytest unit and integration test coverage.

US Bank — Senior AI / ML Engineer (June 2024 – May 2025)

- Engineered a multi-model orchestration engine utilizing Python, LangChain, GPT-4, and Claude Sonnet to automate regulatory compliance verification across millions of records, raising classification accuracy to 95%+.
- Constructed a low-latency hybrid RAG topology combining FAISS, Pinecone, and sparse keyword search, dropping average query response latency from 3.2s to 1.3s while elevating retrieval precision.
- Implemented custom token optimization rulebooks and Redis semantic caching layers for high-volume inference pipelines, achieving a 35% net monthly reduction in commercial LLM API expenditure.

- Constructed Java Spring Boot API Gateway services and low-latency hybrid RAG topologies (FAISS, Pinecone), dropping average query response latency from 3.2s to 1.3s while elevating retrieval precision under sub-second SLA constraints.
- Integrated distributed messaging fabrics via Apache Kafka and FastAPI microservices to interface core banking data warehouses with streaming AI pipelines under sub-second latency constraints.
- Built dual-tier content safety guardrails leveraging Azure Content Safety APIs and downstream PyTorch classifiers, mitigating false-positive security threat alerts by 40% while enforcing zero-trust RBAC controls.
- Established real-time model telemetry, data drift tracking, and experiment logging via MLflow and Weights & Biases, maintaining 85%+ test coverage through automated Pytest suites.
- Scaled serverless feature generation pipelines across AWS Lambda, S3, and SageMaker, accelerating offline vector embedding workloads while reducing cloud infrastructure overhead by 30%.

The University of Kansas — AI / ML Engineer (December 2023 – June 2024)

- Designed high-throughput NLP ingestion pipelines using Python, Pandas, NumPy, and Sentence-Transformers to clean, tokenize, and process 350K+ unstructured research papers, boosting system throughput by 55%.
- Engineered a hybrid retrieval engine fusing dense vector representations with BM25 sparse keyword scores, dropping query noise by 42% and elevating retrieval precision from 68% to 87%.
- Developed dynamic context window compression and chunking routines in Python to eliminate memory overhead spikes, accelerating tokenization throughput during deep document indexing.
- Architected serverless execution pipelines on AWS Lambda and S3 while standardizing containerized testbeds via Docker and Pytest, reducing compute overhead by 30% and accelerating release validation speed.

Cognizant — Software Engineer / ML Engineer (July 2021 – July 2023)

- Developed custom BERT and RoBERTa Transformer configurations for domain-specific Named Entity Recognition (NER) tasks, realizing an 89% F1-score across specialized commercial document parsing workflows.
- Architected distributed big-data extraction pipelines using Apache Spark and Databricks, processing 1.2TB+ of unstructured digital data daily to secure a 3x processing throughput boost.
- Built high-volume text classification microservices in Python and FastAPI to evaluate 3M+ financial files (KYC, loans) monthly at a 91% categorization accuracy rate.
- Engineered containerized deployment pipelines using Docker, Jenkins CI/CD, and optimized PostgreSQL indexing, cutting release timelines from 2 days down to 4 hours while resolving connection pool starvation under heavy read workloads.

Projects

HealthGuard AI – Real-Time Healthcare Monitoring & Compliance Platform

- Developed Java Spring Boot microservices and AI-powered rule engines to monitor healthcare workflows, improving compliance event detection accuracy and reducing manual review efforts by 30%.
- Architected scalable cloud-native solutions using AWS Lambda, SQS, DynamoDB, and API Gateway, supporting high-volume healthcare transactions with high availability and fault tolerance.
- Developed ReactJS dashboards and implemented LLM-powered knowledge retrieval using RAG techniques, improving operational visibility and accelerating access to healthcare policies and documentation.

InsightCampus – AI-Powered Academic Analytics & Reporting Platform

- Developed a full-stack academic analytics platform using Java Spring Boot, ReactJS, and PostgreSQL to streamline reporting and provide actionable insights for students, faculty, and administrators.
- Engineered Python and Apache Airflow ETL pipelines to consolidate data from multiple university systems, reducing report preparation time by over 60% and improving data accuracy.
- Integrated LLM-powered search and RAG-based knowledge retrieval capabilities, enabling users to quickly access academic resources, policies, and institutional information through a unified interface.